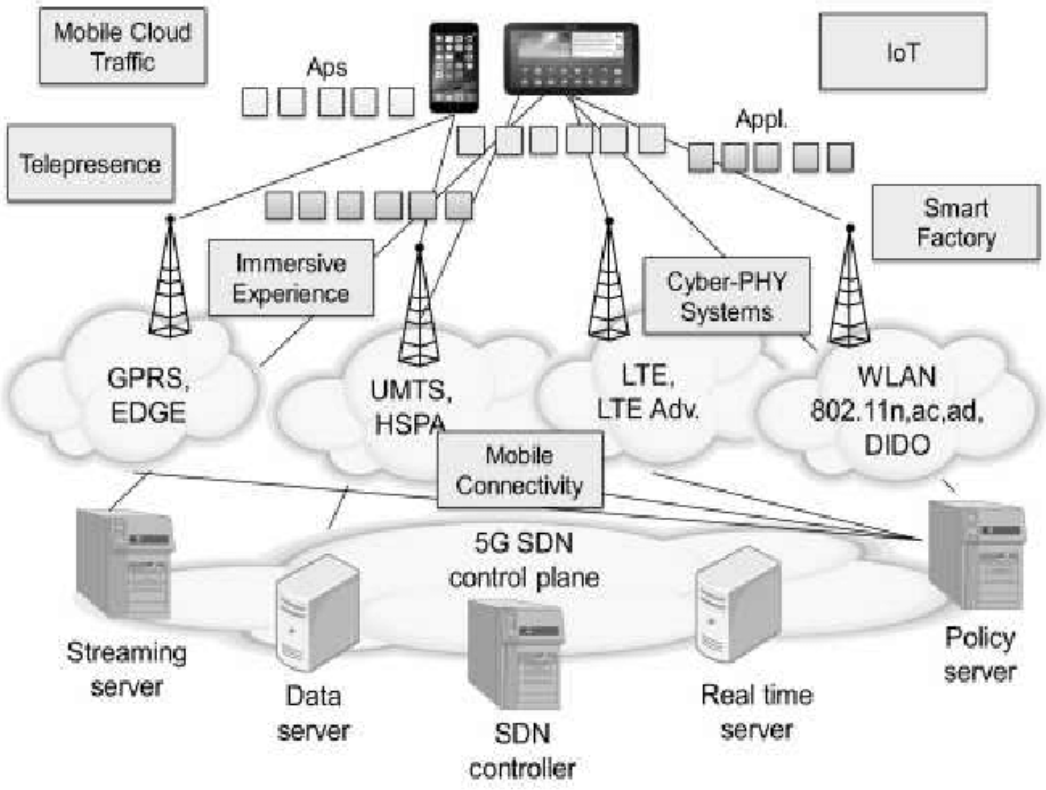
Indian Institute of Information Technology and Management, Gwalior

WIRELESS COMMUNICATIONS TECHNOLOGY ASSIGNMENT

विश्वजीवनामृतं ज्ञानम्

5G COMMUNICATION



Submitted by:

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Introduction

5G system expands the 4G environment by adding new radio(NR) capabilities, but doing so in such a way that LTE and NR can evolve in complementary ways.

5G system entails devices connected to a 5G **Access** networks , which in turn is connected to 5G **Core** network(which offers major improvements over 4G system in the area of **Network Slicing** (N.A.A.S (Network as a service)) and **Service Based Architecture(S.B.A.)**

It's needed because according to current analysis by scientist by data mining and other forecasting techniques , there will be huge increase in mobile users in upcoming future, 1000 times more traffic volume In next decade.

5G cellular System is enabled by millimeter wave spectrum to support high data rates .However Millimeter wave solution impose a requirement for small cells(which is key importance for taking advantage of limited and strategic resources e.g. radio wave spectrum)(cells can be located close to each other without causing interference)(may require many more antennas and antenna sites to overcome weakness and new technologies like Massive MIMO and Beamforming will be the solution)(may also be used in tandem with low frequencies). Approach fostered by 5G std development bodies are focused on providing Super fast mobile network , where density clustered small cells provide contiguous urban coverage to mobile/stationary users. A goal of 5G network is to be 5x faster than

the h i g h e s t current speed of existing 4G network(1 G.bps) i.e. download speeds of 5Gbps [EDITING NEEDED]

5G Objective as per METIS(Mobile and wireless communication Enablers for the Twenty twenty(2020) Information Society) : 1000x Mobile Data Volume per area, 10x to 100x Number of Devices and

User Data Rate, 10x Battery Life for low power I.O.T. devices, more than current system. 5x less end to end Latency than current systems.

5G is expected to support: Tight **Latency**, **Availability**, **Reliability** requirements to facilitate applications related to Video Delivery , Health Care, Surveillance, and Physical Security , Logistics, Automotive Locomotion, and Mission Critical Control, among others, (particularly in an IoT context), A panoply of **Data Rates** (upto multiple Gbps, Mbps) to facilitate existing and evolving applications(particularly in an IoT context), **Network Scalability** and **Cost Effectiveness** to support *Clustured* users with very **high** data rate requirements as well as large number of distributed devices with low complexity and limited power resource.(particularly in an IoT context)(where a rapid increase in the number of connected devices is anticipated)

5G system seeks to satisfy Immersivity , Intelligence, Omni Presence and Autonomy. 5G systems subsume important 4G system concepts (e.g. The energy saving capabilities of Narrow Band I.O.T (N.B.I.O.T) , secure ,low latency (a requirement for autonomous vehicles), small data transmission for low power devices)

Smart City applications lie more within E.m.BB domain . Everyone needs to be connected with the cloud

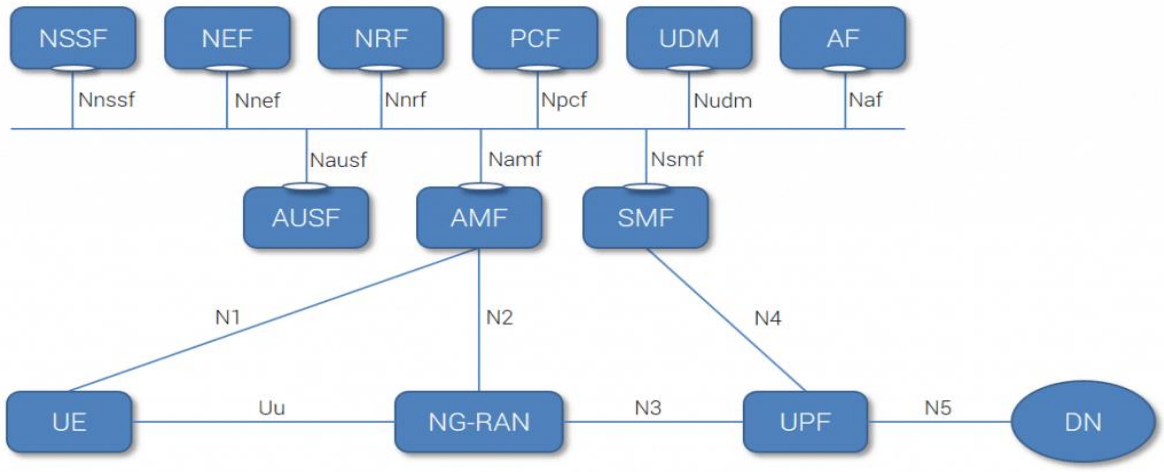
In general Augmented Reality(Augmentation of information in real world) and Virtual Reality applications are needed in every aspect of city operation architecture in which IoT and 5G networks play a key role. AR/VR applications require low latency and high/massive data transmission rate which is possible through : Software Defined Network (S.D.N.) and Multi Path Cooperative Route (MCR) scheme .

5G is a key wireless technology applicable to IOT applications. With Distributed Antenna System 5G can be used indoor (to account for building penetration losses) also. Outdoor use might extend up to hundreds of miles. 5G is evolving and not yet widely deployed. Several bands are allocated for 5G. 5G is a licensed spectrum .Even if implemented with data speeds more than 11 kbps ,battery life may last more than ten years. 5G is cost effective. 5G has entered in some developed countries and before 2025 was expected to be available worldwide. IOT applications and Software based solutions enabled by 5G Communications are and will be boon in crisis situations like COVID-19 Pandemic

5G is supposed to have a peak speed of 5-15Gbps and uses 3 spectrum bands instead 2whereas 4G peak speed is 10-15Mbps, network latency expected to drop to 1ms from 30ms, thus Ideal for IoT

THE 5G REFERENCE ARCHITECTURE. (courtesy of 3GPP)

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UE: (user equipment):User mobile phone;

RAN: (radio access network): access network function which provides connectivity to mobile phone

UPF: (User plane function) ; Handles user plane traffic e.g. traffic routing and forwarding, traffic inspection and usage reporting. Can be deployed in various configurations and locations depending on service.

DN(Data Network); represents operator or 3rd party services or internet The User plane consists of following functions:

AMF(access and mobility management function); performs access , mobility control, transparent proxy for routing SM messages.

AUSF(authentic server function); provides authentication functions UDM(unified data management); stores subscriber data and profiles. It has equivalent role as HSS in 4G, but will be used for both fixed and mobile access in 5G core.

SMF(Session management function); sets up and manages PDU session according to network policy.

NSSF(network slice selection function): selects NSI(NS instance), determines allowed NSSAI(NSS assistance information), AMF set to serve UE
NEF(network exposure function): exposes services and capabilities provided by 3GPP network functions

NRF: (NF repository function): maintains NF profiles and support service discovery.

PCF(policy control functions): provides a policy framework incorporating network slicing, roaming and mobility management

AF(application function): interacts with 3GPP core network(CN) to provide services

Detailed study of used 5G communication frequency spectrum

5G carries information wirelessly through electromagnetic spectrum, specifically the radio spectrum. Within radio spectrum are varying level of frequency band out of which some are used by 5G data accordingly by companies.

Millimeter waves are popular choice of 5G but also has application in areas like *radio astronomy* (analyzing radiation of heavenly bodies) , *telecommunications* (transmission of signal of any nature by electromagnetic or non electromagnetic system(wire,…)), *radar guns* (device used to measure speed of moving objects). Wavelengths in and around 30-300GHz band(part of Extremely High Frequency(EHF) band) often called millimeter band(because wavelengths range from 1-10 mm) and useful in densely packed networks (but not for carrying long distances due to attenuation). Thus one can diversify and use lower bands

UHF (Ultra High Frequency)(.3-3GHz)[which is lower on spectrum than EHF] is also used for 5G and also has application in TV Broadcasting, G.P.S.(Global positioning System) to Wi-Fi, cordless phones, and Bluetooth

Frequency used by 5G tower determines 5G power and data speed, as nature of interaction of wave depends on type of wave (e.g. higher speed and shorter distances for higher frequencies and lower speed and larger distances for lower frequencies [since smaller waveform gets easily distorted by obstacles in path { e.g. use of S.O.N.A.R. in underwater} AND since moving upwards spectrum ,bandwidth increases, thus throughput increases thus data speed increases])

One needs to use an optimum frequency band according to community to balance between trade of high data rate(e.g. demanded by popular cities like Mumbai) and to have services available to large number of devices from single tower irrespective of tower placement(e.g. rural areas)

The industrial consortium setting standard for 5G networks Is 3GPP (3rd generation partnership projects). The air interface defined by it is NR(new radio) and specification classified in 2 frequency bands:

FR(Frequency Range)1[< 6GHz]:

[MAX channel bandwidth: 100MHz](scarcity of continuous spectrum in this crowded FR)

Coverage areas: <2GHz (e.g. .7GHz [indoor and broader coverage area])

C-band: 2-6GHz for coverage and capacity(3.3-4.2GHz[most widely used])

FR2(mmW)[>24GHz]:

channel bandwidth: MIN: 50MHz, MAX: 400MHz

Super-data-layer: > 6GHz(e.g. 24-29 GHz, 37-43 GHz) for high bandwidth areas

Carriers using 5G Spectrum:

T-Mobile (use low-band spectrum(.6GHz),2.5GHz) ,

Verizon(5G ultrawide band(mm waves(28-39GHz))),

AT & T(mm wave spectrum for dense areas and mid and low spectrum for rural and suburban locations),

Sprint(claiming to have most number of frequency bands(.8, 1.9, 2.5)GHz, (in U.S.)

NOTE: 5G spectrum has to be sold or licensed to operators, like through auctions, in order for any company to use a specific band. The international telecommunications union(ITU) regulates the use of the radio spectrum around the world, and domestic use is controlled by different regulatory bodies, such as the FCC in U.S.)

For ensuring wide service,5G networks operate on 3 frequency bands { *low* (.6-.7GHz with download speeds(30-250Mbps),slightly higher than 4G LTE(20-150Mbps)),…), *mid* (2.5-3.7GHz,speed: 100-900Mbps,each cell tower providing service to several mile radius, most widely deployed), *high* (25-39GHz (near bottom of mmW band)

and hence 3 different type of cells

NOTE: As well as acquiring additional spectrum, it's desirable for networks

5G Radiation related health problems

The scientific consensus: “5G tech is safe”. Misunderstanding generated Conspiracy theories have claimed adverse affect(cancer risks and DNA damage, oxidative damage that can cause premature aging, increase in harmful free radicals, …) on human health. Exposure to high levels(> 4watt/kg) of RF-EMF radiation can heat up tissue/raise body temperature and cause sleep disturbances, including insomnia, headache, dizziness, etc. .

A lot of research is being carried out in different countries which can opt 5G currently. Till now infrastructure satisfies WHO RF-EMF international radiation level standards and thus no significant thermal health issues can be related to RF-EMF radiation.

Some studies e.g. in 2011 by IARC(International Agency of Research and Cancer) and in 2015 raise questions regarding carcinogenic effect of RF-EMF and increased risks of glioma(type of intracranial tumour that occurs in brain and spinal cord) and acoustic neuroma(a non cancerous tumour on the main nerve leading from the inner ear to the brain) in heavy users of mobile phones(>= 10 years)

Apart from carcinogenic activity of RF-EMF , few epidemiological studies on subjective ailments(e.g. headaches, fatigue, sleep disturbances, negative impact on cognitive cycle and memory, affect reproductive system and how it functions, effect on communication between neural and mascular system(esp. by mmW)) caused by both hand-held telephones and bases stations (RF-EMF sources)

Cell phone network root concern is non ionizing radiofrequency radiation(RFR) which according to some studies, when in excess(high

exposure)[e.g. to 3G] may cause tumors in {heart, ..}(cancerous), or in adrenal gland of male rats. 5G technology is a low level hazard but possesses no risk. 5G technology possesses low level hazard not only due to presence of mmW but also because of increase in cells(which may be located at street lights, electrical poles,…) used to channel it .

Thus since 5G is inevitable , To minimize health risk we can Unplug/Turn Off cell phone/wi-fi when not in use, don’t store cell phone close to body etc…

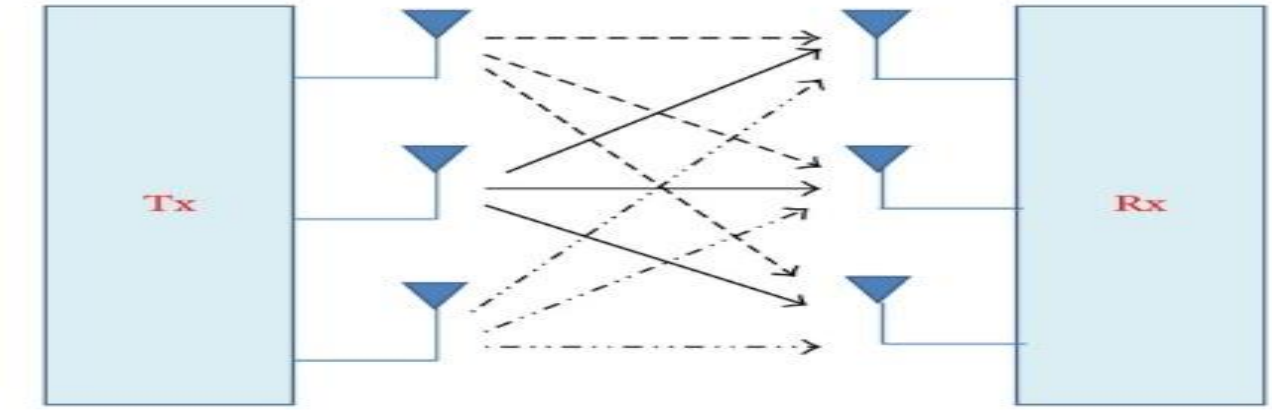
MIMO antenna, its characteristics and applications explained

MIMO Antenna:

MIMO has become essential part of wireless communications using WIFI, IEEE 802.11ac(WIFI), HSPA+(3G),Wi max and LTE(long term evolution) has revolutionized high speed wireless. It was discovered by "Arogya swami Pau l raj"

MIMO(Multiple input Multiple Output) systems/antenna are traditionally defined as point to point communication links with multiple antennas at both the transmitter(Tx) and receiver(Rx) for multiplying the capacity of radio link/channel(both in theory and practice) and to minimize errors and optimize data speed. But in modern times MIMO refers to practical technique for sending and receiving multiple data signals at both transmitter and receiver simultaneously over the same channel.

Hence its better than "diversity" systems which have multiple antennas at Tx or Rx and SISO systems in general. In particular research has shown MIMO systems can significantly increase the data rates of wireless systems without increasing transmit power or bandwidth at expense of added cost of deploying multiple antennas , space requirements of these extra antennas (esp. on small handheld units), and added complexity required for multidimensional signal processing. Recent work in MIMO includes capacity of MIMO systems under different assumptions about channel knowledge , optimal coding/decoding, transmission strategies for un coded systems



A narrow band flat fading model: Y(y1,…y(m))T = H[h(i)(j)](m*n) * X[x1,…,x(n)] + N[n1,..,nm] (may be added white gaussian noise(AWGN))

MIMO characteristics:

MIMO can be subdivided in 3 MAIN categories. Precoding, spatial multiplexing (SM) , AND diversity coding.

Precoding In general terms is considered to be all spatial processing occurring in transmitter AND multi stream beamforming in particular and is required since precoding with multiple streams is often more beneficial than SINGLE STREAM transmit beamforming(traditional unpractical approach in cellular networks of today ,which are mainly categorized by multipath propagation. where the same signal is transmitted from each of transmit antennas with appropriate phase lag and weightage gain (AND well defined directional pattern IN (line of sight)LOS propagation) so as to maximize power at receiver input by constructive addition of signals from different antennas and minimize multipath fading effect.) for maximizing signal power at all of receiving antennas simultaneously But it REQUIRES INFORMATION OF CHANNEL STATE INFORMATION(CSI) at transmitter n receiver.

. Spatial multiplexing(SM) requires MIMO antenna configuration and is very powerful technique for improving channel capacity at higher SNR(without CSI AT Transmitter, But can be combined with pre coding if CSI is available.). SM can be used for simultaneous transmission to multiple receiver. known as space division multiple access or multi user MIMO(MU-MIMO). In which case CSI IS REQUIRED BY THE TRANSIMITTER.

In SM, a high rate signal is split into multiple lower rate streams(<=MIN(n ,m)) , and each stream is transmitted from a different

transmit antenna in same frequency channel . if transmitted signal arrive at the receiver antenna array with sufficiently different spatial signatures and the receiver has accurate CSI. It can separate these streams into almost parallel channels.

The scheduling of receivers with different spatial signatures allows good separation.

Diversity coding techniques exploits independent fading in the multiple antenna links to enhance signal diversity and are used when there is no channel knowledge at the transmitter(hence there is no beam forming or array gain) and Can be combined with spatial multiplexing, when some channel knowledge is available at receiver. In diversity methods, a single stream (unlike multiple stream in SPATIAL multiplexing) is transmitted but signal is coded with techniques like space time coding and is emitted from each of the transmit antennas with full or near orthogonal coding

MIMO exists in many versions like

COOPERATIVE MIMO (CO-MIMO) :

Which Uses multiple neighboring BS(base stations) to jointly transmit/receive signals from users. As a result, no intercell interference issue (which is in conventional MIMO systems)

Massive MIMO :

A technology where n(terminals)<<n(MS(mobile station)/BS antennas). Benefits exploited In rich scattering environment in accurate perfect-CSI using simple beamforming strategies: maximum ratio transmission (MRT), maximum ratio combining, or zero

forcing(ZF). However in practice, channel between transmitter and receiver is estimated from orthogonal sequences which are limited by coherence time of the channel. Most importantly, simple pilot assignment and channel estimation methods are used(discovered from limited training sequences from proposed works.) in a multicell setup since the reuse of pilot cell sequences of several co channel cells will create pilot contamination steeply degrading MIMO performance. Thus Massive MIMO is an extension of MU-MIMO , in which base station transmitter simultaneously communicate with multiple mobile stations receiver using same time frequency resources, improving spectrum efficiency

MULTI USER MIMO(MU-MIMO)

In recent 3GPP and WIMAX std, MU-MIMO is being treated as one of the candidate technologies adopted in the specification of number of companies including Samsung, intel ,QUALCOM, Ericson, Huawei, Philips, Nokia, Freescale. For these and other firms active in mobile hardware market, MU-MIMO is more feasible for low complexity cell phones with a small number of reception antennas. Where as single user SU-MIMO's higher per-user throughput is better suited to more complex user devices with more antennas.

Enhanced multiuser MIMO: 1) Employs advanced de and precoding techniques

SDMA: space/super division multiple access is used. Where super => non orthogonal techniques: superposition coding used; instead of orthogonal division: time/frequency division .

MIMO Applications

MIMO is planned to be used in Mobile Radio Telephone standards such as recent 3GPP and 3GPP2. In 3GPP, High speed Packet Access plus(HSPA+) and LTE standards take MIMO into account. Moreover, to fully support cellular environments, MIMO research consortia including IST-MOSCOT planned to develop advanced MIMO techniques(MU-MIMO(multiuser MIMO))

MIMO tech use : non wireless systems. E.g. (home n/w) ITU-TG.9963 which defines a powerline communication system that uses MIMO technique to transmit multiple signals, multiple AC wires(phase, neutral and ground)

The IEEE 802.11e standard incorporates MIMO-OFDMA with spatial multiplexing(SM). The IEEE 802.11n standard , released in October 2009 recommends MIMO-OFDM with SM. Since SM make receivers very complex, and therefore they are typically combined with Orthogonal Frequency Division Multiplexing(OFDM) or Orthogonal Frequency Division Multiple Access(OFDMA) modulation, where problems created by a multipath channel are handled efficiently

Short out problems during installation of future 5G network

Technical and Business challenges:

1) spectrum 5G network deployment challenges

5G NR deployments in low(e.g. .6, .7, 2.6, 3.5 GHz) and high(e.g. Mm W : 15,24, 28,39 GHz) frequencies , comes with various scopes and challenges

Sub 6GHz spectrum requires low site density to ensure delivery of blanket mobile connectivity which comes at expense of narrower available bandwidths(generally 100MHz, using up to 4 carrier aggregation, CA) and consequently, peak throughputs of Gbps, order of magnitude, therefore higher order modulation schemes and MMIMO optimization are crucial, for delivering high quality consumer, EMBB applications , such as HD video streaming, and MMTC applications like smart cities.

It's worth mentioning T-Mobile's approach in the United States, with their 5G Deployments of 600MHz blanket coverage, for smooth and cost efficient mobile 5G connectivity.

In another example, due to penetration and propagation loss, deployments in the higher spectrum, like Sprint's 2.6GHz solution, come with more 5G network challenges than sub 1-GHz spectrum.

High frequency MMW spectrum, characterized by available bandwidths of at least 1GHz ,require high site density

This spectrum is suitable for hotspots, such as high traffic and population density, and for enterprises wither higher bandwidth

demanding use cases, such as EMBB for UHD, 4k, 8K AND 3D video streaming, and remote diagnostics for healthcare

MMW can also be applied to URLLC, for low latency demanding applications, such as augmented reality, virtual reality, remote surgery mentoring, and industrial IOT applications..

However, MMW 5g deployments have significant challenges from propagation characteristics and dependencies, like vegetation , building materials and weather.

These deployments allow high capacity for only short distances, while jeopardizing the delivery of consistent coverage.

Although , during non LOS trials, the environment demonstrated better coverage than expected, due to reflections.

The first 5G deployment must evaluate how well devices pick up and benefit from MMW reflections.

Fixed wireless access(FWA) deployments can benefit from LOS, but only if mounted on rooftops that are approximately 40m high.

Planning and evaluating coverage range loss in various antenna mounting scenarios becomes critical for FWA use cases

Additionally 5G deployments of MMW high density sites require automation of processes and testing scenarios

It’s noteworthy that MMW spectrum is used in 5G deployments for FWA In the Verizon 28GHz solution, and very low(walking) mobility by AT&T, using 39GHz with a puck device

2) MMIMO /3D beamforming challenges

Among all the 5G technology disruptions, MMW,MMIMO ,3D ,flexible numerology and slot structure; MMIMO and 3D beamforming have the most improved spectral efficiency

Multiple access with pinpoint beam forming, allows multiple users to share the same spectrum.

This ensures increased capacity and coverage, especially at the cell fringe,

Additionally, 5G deployments In sub-6GHz can benefit from pre-5G MMIMO solutions deployed at LTE sites, by co-siting NR and LTE.

This can reduce cost and ease the technical difficulty of MMIMO Implementations.

However, the first 5G deployments are expected to face performance challenges, because of the dependency of channel estimation, multi user multi stream, optimization algorithms, and beam management proprietary solutions, that are within devices and vendor equipment.

Quantification of MMIMO gains and 3D beam distribution and usage, requires device based evaluation, to understand the end impact of user experience.

3) Hybrid LTE-NR and Mobility Challenges,

The first 5G deployments are non-standalone(NSA) solutions.

These deployments use LTE core, while LTE access represents the anchor for NR.

LTE ENB ensures the control, while 5G GNB provides additional user traffic on top of the LTE traffic

Cost efficient sub-6GHz co-sited ENB and GNB can smoothly ensure significant coverage and throughput increase

5G deployments in MMW spectrum, can benefit from the hybrid LTE-NR configuration by using the macro-LTE layer as a coverage blanket for MMW mobility between hotspots.

This MMW mobility has proven to work at low speeds (walking), while high speed mobility still requires extensive tests and evaluations.

However, these benefits are challenged by the complexity that emerges from the performance of the dual connectivity mode(EN-DC,E-UTRAN-NR dual connectivity).

This requires precise synchronization between ENB and GNB for signaling and data transfer to the NR

4) ***Throughput and latency performance challenges***

   Spectrum allocated bandwidths and spectrum types, such as EMBB -enabled streaming, AR and VR, define the targeted performance.

   Each service types experiences different conditions and requires different testing scenarios, to understand how throughput and latency impact coverage conditions and configurations

5) ***Device challenges***

The timing of initial 5G deployment is determined by 5G device availability, which is still scarce.

This is due to several technical challenges, including multi-band support of upper and lower frequency bands, that cause design challenges for the front-end filters

It also poses heating concerns, because of power consumption needed to transmit in high frequency bands, and a significant performance impact for higher data rates and bandwidths

MMIMO antenna positioning determines beamforming gain/loss and beam management in the devices.

This has immediate impact on device coverage and mobility performance .

5g deployments has extremely complex device based testing requirements, with a large number for scenario combinations needed to understand and quantify device performance and device interaction with the network

6) 5G uses MMW which are signals(which follow LOS[which is impossible again]) of high frequencies(for transmitting at higher speed) that can be blocked by our body, trees or other obstacles and can be faded few km away from source before reaching => increase density of cellular base stations in an area ,but practically it's impossible to have cellular BS every km=> engineers looking at placing small signal cells after every dozen electricity poles in a neighborhood, and initially this service will not be available to entire mass as a mobile operator needs to invest a lot of money to

insert 10 times number of transmission cell in one particular area(which in extensive use(city))(as in 4G LTE). The high speed data transmission rates will cause a huge battery drain in phones running 5G network=>risk accidents: battery burst/overheating/catching fire=> thus new generation battery needed .

7) Download speed (<=1.9Gbps) >> upload speed(.1Gbps)

Possibility of home automation during 5G communication.

Based on Network Slicing concept ,which makes use of network function virtualization(NFV) and software defined network(SDN).

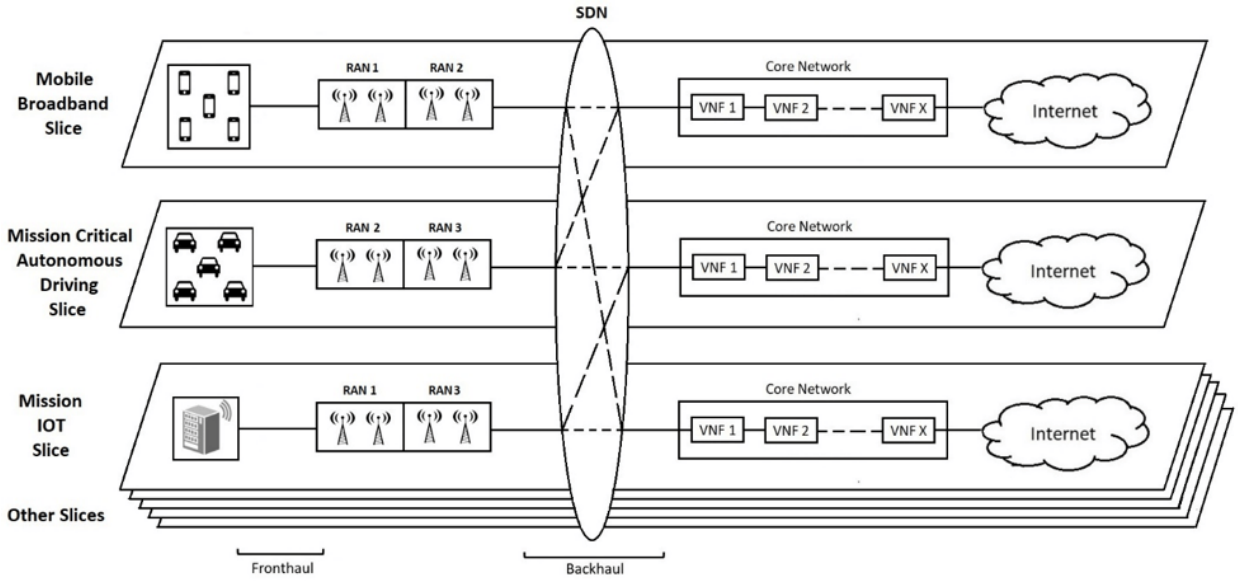
Smart(connected/automated/) home/ Dom o tics: an INTELLIGENT home which provides homeowners security(smart locks, security cameras), comfort(smart TV, blinds, thermostats), convenience(smart lightning, appliances, irrigation), and energy efficient that contains a communication network that CONNECTS different appliance and equipped with lightning, heating, and electronic, devices that can be CONTROLLED remotely by smart home app on their smartphone/computer/other networked device (e.g. u can contact smartphone on internet to make sure if food is cooked, A.C. is on, enough milk is there(say)…)

In order to function properly ,these smart devices need to be connected properly for which The first communication protocol X10 was developed in 1975 by Pico electronics of "Glen r o the s", Scotland, with objective of allowing remote control of home device and appliance. And recently Zigbee and Z-wave are 2 most used home automation communication protocols, which are short range and low power wireless technologies. To establish home network it's necessary to connect router to modem.

5G network slicing is a composition of adequately configured networked functions, network applications, and the underlying cloud

infrastructure(physical, virtual or even emulated resources, RAN resources) that are bundled together to meet the requirements of a specific use case, e.g. bandwidth, latency, processing, and resiliency, coupled with a business purpose.

Generally 3 network slices are proposed: (fig)



However, there could be smart security system network slice above the (enhanced) mobile broadband slice.

Although 5G cannot be a panacea for smart home due to battery life.